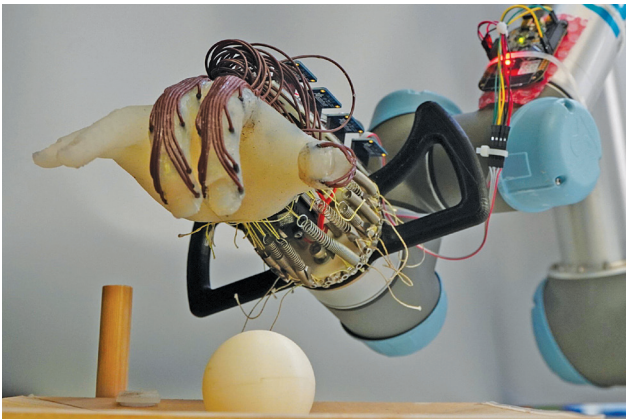




Bochum, are exploring methods to detect hardware Trojans by comparing chip blueprints with electron microscope images, successfully detecting deviations in 37 out of 40 cases. Researchers analysed chips in modern technology sizes and couldn't modify them due to built-in hardware Trojans. Instead, they retroactively changed designs to create deviations between construction plans and chips. They then tested the ability to detect changes without prior knowledge. First, researchers prepared chips using chemical and mechanical methods to take thousands of images of lowest chip layers, which contain several hundred thousand standard cells that perform logical operations. Modifications were successfully detected in chips of 90, 65, and 40 nanometers with 500 false positives. However, the detection process became difficult for 28 nanometers due to impurities blocking rows of standard cells.

Low-cost, 3D-printed soft robotic hand can grasp objects and avoids dropping

Researchers from the University of Cambridge have designed a low-cost, energy-efficient, 3D-printed soft robotic hand that can grasp objects and avoid dropping using wrist movement and 'skin' sensation. The robot's sensor-equipped hand can

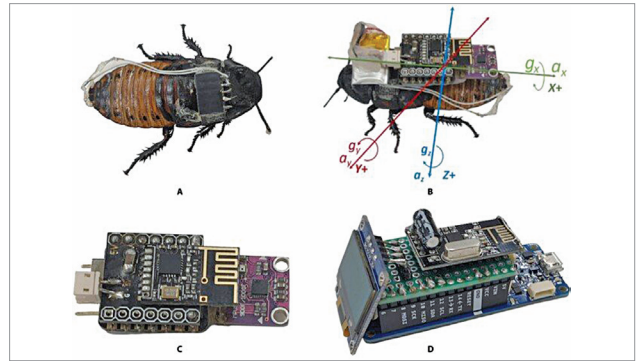


Researchers have designed a low-cost, energy-efficient robotic hand that can grasp a range of objects—and not drop them—using just the movement of its wrist and the feeling in its skin (Credit: University of Cambridge)

anticipate object dropping. Researchers have created a 3D-printed hand with tactile sensors and passive wrist movement. They conducted 1,200 tests on the hand's gripping ability. The robot hand learned from humans using plastic balls. It can lift without finger actuation, and tactile sensors predict grip effectiveness and potential failures. The robot tested successful grips with 14 objects, sensing pressure with its 'skin' sensors. It learns from failures and customizes its solution. The hand has a single strategy for picking up many items and is actuator-free, simplifying design.

Converting a cockroach into a cyborg being explored for search and rescue missions

Scientists at Osaka University have proposed the idea of converting a cockroach into a cyborg. Scientists believe that cyborg cockroaches, with superior mobility, are ideal



Scientists from Osaka University designed a cyborg cockroach and optimised its movement by utilising machine learning based automatic stimulation (Credit: Cyborg and Bionic Systems)

for search and rescue missions. However, natural cockroach behaviour and low-light monitoring pose challenges. This study aims to improve cyborg cockroach movement. Machine learning can detect cyborg cockroach motion state via IMU and electrically stimulate their brains for movement in darkness or cool environments. The online detector minimizes overstimulation, providing autonomy. Cyborgs showed better search rates and longer travel distances while reducing stop time. Stimulating the cerci of Madagascar hissing cockroaches induced free-walking motion, overcoming their reluctance to move in dark and cold environments.

Can a robot help you eat your meals?

A research team in Dorsa Sadigh's ILIAD lab aims to improve robot-assisted feeding. They created new robotic algorithms to assist with feeding different food types. One algorithm uses vision and haptics to insert the fork correctly; another uses a second robotic arm to push food on a spoon; and the third algorithm delivers food comfortably to the mouth. Researchers used a camera and force sensor to teach a robot arm how to handle different food textures. The robot uses prior knowledge to approach food and detect fragility, choosing between two skewering methods. Researchers used a bimanual robot to handle various foods, replacing the traditional method. A vision system detects food fragility and adjusts utensils to prevent breakage. Researchers made a robotic feeding system with human-like movement, computer vision, and force sensing. It detects when a person eats and can switch controllers and reactivity levels.